Chapter 8

Appendix

Differences between evaluators

In this online appendix, we discuss some of the differences in the methods used by the four evaluators and provide short summaries of each evaluation. The appendix is more technical and likely to be less accessible to the lay reader than the main text.

The four evaluators are Krekel and colleagues, State of Life, Pro Bono Economics, and the Happier Lives Institute (HLI; where the authors work). There are several notable differences between how the evaluators assess their charities. Our comments here focus in large part on the differences between HLI and the other evaluators. We do so for two reasons. First, we know HLI's analyses better. Second, we feel like it's worth trying to further explain some reasons why HLI's cost-effectiveness numbers are higher.

The first difference is the location of the charities. Krekel and colleagues, State of Life, and Pro Bono Economics focus almost exclusively (except for the Tearfund analysis by State of Life) on charities operating in the United Kingdom, whereas the HLI analyses mostly focus on countries in low- and middle-income countries (LMICs). As we noted in the main text, we think this difference is largely due to the differences in goals each evaluator had. The Happier Lives Institute is trying to find the most cost-effective ways to improve wellbeing, whereas the other evaluators, who are (also) based in the UK, were assessing organisations that were convenient to assess and/or they were asked to assess.

The second notable difference is that there is a large variation in the relevance and causal interpretation afforded from the evidence used in the cost-effectiveness analyses between researchers.

It's an outstanding problem in science to understand how to trade off between relevance (i.e., generalisability or external validity concerns) and the quality of the causal evidence (i.e., internal validity concerns). Our confidence should be highest when an evaluation is supported by high-quality causal evidence (i.e., multiple large RCTs), *and* it's highly relevant, reflecting the programme's particularities in practice. But this is hard to come by. A notable exception is HLI's evaluation of GiveDirectly which used several large, high-quality RCTs of the programme as it is deployed in practice.

Causal interpretation of evaluator's evidence

HLI's evaluations typically draw on natural experiments or RCTs which afford a stronger causal interpretation of the relationship between the intervention and the effect on wellbeing. In contrast to the non-HLI evaluations we reviewed, only the Frijters and Krekel evaluation of Action For Happiness and the Krekel et al. analysis of the Royal Voluntary Service used causal evidence as the basis of their analysis.

Pro Bono Economics evaluations use matching estimators combined with a difference-in-difference analysis, which does not afford as much causal interpretation as RCTs but is better than using purely associative data like pre-post results. State of Life has also used some methods which, while affording less causal interpretation than the methods above, are better than purely associative estimation strategies. For Parkrun, they apply adjustments for the counterfactuals. For Tearfund, there is no randomisation but they do compare results between participants and people not participating in the programme The rest of the evaluations used other non-causal estimation strategies. These included looking at the association between participation in the programme and wellbeing or looking at the change in wellbeing before and after participation but without a randomised control group for comparison.

Relevance of evaluator's evidence

Non-HLI analyses tend to rely almost exclusively on directly relevant (but mostly non-causal) evidence. By "relevance" we mean the degree to which the evidence relates to the programme a charity deploys as it's delivered in practice. For example, Pro Bono Economics bases its analysis of Football Beyond Borders on a dataset that includes participants directly affected by the programme as it is deployed in practice.

HLI's evidence comes from a mix of more and less relevant sources, incorporating direct evidence when available. For example, in the HLI analysis of StrongMinds and Friendship Bench, we used a meta-analysis of randomised control trials (RCTs) of psychotherapy in LMICs in general as well as direct evidence (RCTs of the programmes of the charities and pre-post data from the charities' monitoring and evaluation). Note that HLI makes use of external validity adjustments to compensate for the limitations in relevance of some of the data we use.

There are also some further differences between analyses that we have identified, although these may not be the only ones. These differences raise some concerns about whether we can compare across these evaluations. Some factors mean the HLI analyses should be more conservative, and others more generous in terms of estimating the cost per WELLBY. We attempt to categorise these differences depending on whether they seem to lean towards HLI under- or over-estimating cost-effectiveness relative to other evaluators.

Factors leading towards HLI being relatively more conservative:

- We attempt to base our estimates on causal evidence. In our experience, causal estimates often seem to be smaller than effects based on pre-post or other associative estimates.
- We assume any cost not going to the evaluated programmes for a charity has no effect, where some of the other analyses only look at the costs of a particular programme (when a charity delivers several).
- We consistently apply adjustments to account for potential issues with internal or external validity. These generally lead to estimating the true effect is smaller than that suggested by the evidence.

Factors leading towards HLI being relatively more generous:

- We include spillover effects such as the effect on those in the household of the recipient.

Factors that could lead towards HLI being either more conservative or generous (unclear):

- We attempt to estimate the duration of the effects and do not just assume they all last a certain amount of time (e.g., one year). This sometimes leads to a longer estimated effect than a year (the typical assumption in the other evaluations) and sometimes shorter.
- Since we use meta-analyses, which standardise effects into standard deviation (SD) changes in the outcome, we need a way to convert SD changes to changes on a 0–10 life-satisfaction scale. We convert SD results into WELLBYs by multiplying them by 2, based on the average SD on the Cantril Ladder in the *World Happiness Report* data.[1] We do this by assuming all changes in measures of subjective wellbeing, distress, depression, and anxiety are equivalent in value as per some of our research.[2] Note that other evaluators sometimes convert from scales with different ranges than 0–10 life-satisfaction scales (e.g., Pro Bono Economics with Football Beyond Borders). For example, they may stretch values derived from a 1–5 scale by a factor of (11/5) to match the range of a 0–10 scale.

In the following sections, we'll provide brief descriptions of the evaluations, broken down by each team. We have already given some of these above in the main text. When we have covered an evaluation already in the main text, we simply restate what we said there, but we include all of them here for completeness. Readers who have read the summaries in the main text should feel free to skip over those.

Table A1: Frijters, Krekel, and colleagues charity WELLBY cost-effectiveness analyses

Charity	Intervention type	Cost per WELLBY	Duration assumed (in years)	Total sample size	Country (income)
Royal Voluntary Service	Volunteering	$81.99	1	4,033	UK (HIC)
Action for Happiness	Happiness courses	$100.00	1	146	UK (HIC)
Restoration Trust: Human Henge	Mental health support	$3,568.93	1	20	UK (HIC)

Krekel and colleagues

As we explained in the main text, we use "Krekel and colleagues" to refer to two academic research teams where one of the lead authors in each case was Christian Krekel. This is a shorthand to refer to the WELLBY analyses discussed in Frijters and Krekel (2021) and Krekel et al. (2024).

The three analyses performed by Krekel and colleagues vary in their methodology, so we leave further discussion of their approach to the relevant sections.

Action for Happiness

Note this description is identical to the one in the main text.

Action for Happiness[3] is a charity that delivers courses and promotes wellbeing skills. The analysis[4] is based on a randomised control trial which found that a six-week course teaching wellbeing skills raised life satisfaction by around 1 point on a 0–10 scale at the two-month follow-up. To get the total effects over time the authors assumed the effects remained constant for a year before dropping off completely. They used the course data that suggests it costs £80 per participant and assumed these costs and randomised control trial results are representative of the charity in general. However, Action for Happiness has since switched from an in-person to a virtual model, meaning the evidence is now less relevant to the delivery in practice. Overall, Action for Happiness has a cost per WELLBY of £80 (or $100)[5]. We consider this to be a shallow analysis.

Royal Voluntary Service

The Royal Voluntary Service[6] is a UK charity dedicated to supporting older and vulnerable individuals by providing volunteer-led services in the community, home, and hospitals. In partnership with the NHS and the UK Department for Health and Social Care, they are rolling out a new volunteer programme. A similar programme was used during the COVID-19 pandemic, where volunteers signed up to check in and chat with isolated people or assist individuals, NHS sites, and pharmacies (e.g., with deliveries). A study[7] of 4,358 volunteers found that the act of volunteering increased the life satisfaction of volunteers by 0.21 points on a 0–10 scale. Volunteers were assigned to treatment if, after signing up, they had already been allocated to and finished a task, and assigned to control if not. The programme cost £3.1 million and yielded 225,069 volunteers in the period of April to July 2020. This leads to a cost per WELLBY of £66 (or $82)[8]. These results are extrapolated to the Royal Voluntary Service

at large. Note that this does not include the benefits in wellbeing to the recipients of the volunteers' help. This analysis is medium in depth.

Human Henge

Human Henge[9] is a UK charity that provides eclectic workshops around art and heritage aimed at improving the wellbeing of people with low mental health (i.e., they call it 'Culture Therapy'). The analysis[10] was based on 10 weekly 3-hour sessions involving eclectic workshops around historical sites like Stonehenge. With a pre-post (not casual) method, they found that the 20 participants had an increase of 0.68 points on a 0–10 life-satisfaction scale[11]. They assumed the effects would last a year. This leads to a total benefit of 13.6 WELLBYs. It cost £38,830 for this programme for 20 people, resulting in a cost per WELLBY of £2,855 (or $3,569)[12]. This analysis is shallow in depth.

Hypothetical charity to help the homeless with housing and other services

Frijters and Krekel (2021) estimate, based on an RCT, that providing "housing first" for people experiencing homelessness is reasonably effective, providing 0.67 WELLBYs per person[13]. However, they also found it was very expensive, costing around $23k per person. Thus, they concluded that providing a housing first intervention costs $35k per WELLBY[14] created.

State of Life

State of Life describes themselves as an organisation that "helps organisations (large and small) to evaluate and measure the social impact and economic value of their activity or project. State of Life's expertise lies in quantitative analysis, particularly in measurement and evaluation of wellbeing outcomes, in line with the 2021 HM Treasury Green Book, in which State of Life are named advisors in the Supplementary Guidance on wellbeing."[15]

At the time of writing, State of Life has conducted and published three cost-effectiveness analyses of charities based on WELLBYs (which we summarise in Table A2 below)[16]. All of State of Life's evaluations are based on relevant (i.e., the people surveyed are recipients of the charity; see more detail in endnote[17]) but non-causal evidence. Typically, their methodology involves a survey of the relevant population where they interpret the relationship between participation in the charitable activity and life satisfaction as causal and lasting a year. They then use the total costs from the charity involved to deliver the particular intervention. Below we provide a brief overview of each evaluation.

Table A2: State of Life WELLBY cost-effectiveness analyses

Charity	Intervention type	$ Cost per WELLBY	Duration assumed (year)	Total sample size	Country income
Tearfund	Multifaceted, religious	$39.33	1	7,212	Several countries in Africa (LMICs)
Parkrun	Exercise, volunteer	$205.67	1	567	UK (HIC)
London Youth Rowing's Active Row	Sport, exercise	$500.00	Unclear	525	UK (HIC)

Tearfund

Tearfund's Church and Community Transformation intervention (CCT) is a flexible multifaceted programme that trains local churches to identify the best way they can benefit their community and implement them with the community (e.g., setting up support and savings groups)[18]. State of Life[19] studies the programme as it is delivered in Rwanda, Sierra Leone, Tanzania and Zimbabwe using a survey of three groups: those who have participated in CCT groups (n = 4,324), those in a community with a CCT church who do not participate (n = 1,236) for spillover purposes, and those in communities without a CCT church (n = 1,767) at present but who Tearfund plans to invite (a sort of waitlist).

This is not a randomised control trial (i.e., it does not establish causality) as communities were not randomly assigned to either participate in the CCT programme or remain on the waitlist. Instead, participation was based on communities accepting an invitation to join the CCT programme. Nonetheless, comparing active communities with those not yet engaged provides more robust insights than solely analysing pre-post results within the active communities. This approach, however, introduces the possibility of selection bias, where participants in the CCT programme may exhibit higher life satisfaction for reasons unrelated to their participation. State of Life addresses this concern in Section 6, arguing that the bias is unlikely to skew results upwards because communities initially accepted into the programme were selected based on greater need, which suggests they may have started with lower baseline life satisfaction compared to those still on the waitlist.

State of Life presents results as a lower and upper estimate, so we take the liberty of averaging between the two to present a central estimate. They find that participating in a CCT programme is associated with an increase of 1.4 points on a 0–10 life satisfaction scale compared to those not in a CCT community. They assume this effect lasts a year. They estimate that one CCT community has, on average, 107 participants a year. They estimate that it costs $5,715 to run a CCT community for one year ($709 direct costs to Tearfund, $1,255 direct value of the volunteer's time, and $3,750 indirect costs from intermediate resources mobilised by the community). This results in a cost per WELLBY of $39. They provide an analysis with spillovers but they do not consider it their main estimate[20]. This analysis is of medium depth.

Parkrun

Note this description is identical to the one in the main text.

Parkrun is a UK charity that supports free community runs delivered by volunteers every weekend. State of Life (2021a) estimates its effects using correlational evidence. First, they use a larger but less relevant dataset from the UK (n = ~60,000) that shows the relationship between running, volunteering, and wellbeing more broadly. Second, they use a longitudinal study that followed participants before and after they participated in a Parkrun event in the UK (n = 576).

They apply these effects to the total number of walks or runs (8,590,393) and volunteering for Parkrun instances (974,255) in 2019 (see their Table 8 and Approach 3). They then discount the effects in both cases to account for the fact they are missing a randomly assigned control group and so can't be sure how much of the correlation is due to the effect of related activities.

They did not provide a central estimate between the two estimates, so we took the liberty of averaging them together. The result is Parkrun producing 27,651 WELLBYs in 2019. The total operational cost of Parkrun in 2019 was £4.5 million, resulting in a cost per WELLBY of £165 ($206)[21]. We consider this analysis to be of medium depth.

London Youth Rowing's 'Active Row'

London Youth Rowing's 'Active Row' programme teaches students in secondary schools how to row[22]. State of Life used a survey of 525 pupils[23] drawn from participating and control schools (not randomly assigned) to estimate that the programme has a cost per WELLBY of £400 (or $500)[24]. No further details of the analysis are publicly available. We suppose that this analysis, like the rest of State of Life's analyses, is of medium depth.

Table A3: Pro Bono Economics wellbeing cost-effectiveness analysis

Charity	Intervention type	Cost per WELLBY	Duration (years)	Total sample size	Country
Walking with the Wounded Head Start	Therapy	$1,674.81	1	118	UK (HIC)
Walking with the Wounded Employment	Employment	$5,601.19	1	92	UK (HIC)
Football Beyond Borders	Sports, education and counselling	$8,690.85	1	153	UK (HIC)

Pro Bono Economics

Pro Bono Economics is a non-profit consultancy that describes themselves as using "economic analysis and the unique insight provided by our connection to the social sector to help charities, funders, firms and policymakers to collectively tackle the causes and consequences of low personal wellbeing in the UK."[25]

They have supported many charities (they report helping 137 in 2023) and performed many CEAs of charities, but only four using WELLBYs. All of the charities they evaluated with a wellbeing lens are based in the UK. Similarly to State of Life, they also rely on non-causal evidence. Nevertheless, in their WELLBY analyses, the methodology is closer to affording a causal interpretation because they create a control group (i.e., a matched control) by statistically identifying individuals with similar characteristics to those receiving the intervention. We summarise the cost-effectiveness of the charities they have evaluated below in Table A3.

Walking with the Wounded

Pro Bono Economics evaluated two programmes delivered by Walking with the Wounded[26]. Walking with the Wounded is a British charity to help injured former British Armed Forces servicemen and women in their career transition from the military to civilian life. One was a therapy programme (Head Start) and the other was an intervention to increase employment[27]. Treatment was defined as completing 12 to 18 sessions of therapy for Head Start or completing 30 days of employment support for the employment intervention.

Both were evaluated by taking the life-satisfaction effects before and after, and then comparing this change to a control group formed by matching each participant to the most similar individuals based on observable characteristics[28], drawn from the Understanding Society Dataset. This approach, known as a matched differences-in-differences estimator, is often taken by researchers to be sufficient for causal inference, but the authors of the PBE report they did not think this was warranted given issues with data quality[29].

They estimate a benefit of 1.33 life-satisfaction points (on a 0–10 scale) for Head Start (therapy), and 0.42 life-satisfaction points (on a 0–10 scale) for the employment programme. Pro Bono Economics do not explicitly do a WELLBY CEA, but they report figures that we can use. The cost per person treated in each programme from 2021 (the latest year mentioned in their report) was £1,782 (Head Start) and £1,882 (employment programme) per person. We assume this effect

lasts a year, which is the assumption used by State of Life as well as Krekel and colleagues in their evaluations. Given these inputs, the cost per WELLBY is £1,340 (or $1,675) for Head Start and £4,481 (or $5,601) for the employment programme[30].

Football Beyond Borders

Note this description is identical to the one in the main text.

Pro Bono Economics evaluated[31] the cost-effectiveness of Football Beyond Borders[32], a UK charity dedicated to improving school outcomes for students through a combination of tutoring with a trusted adult, football practice, and therapeutic support.

The study was not a randomised control trial, but they used a matched control method and a difference-in-difference estimator, which is better than relying on pre-post changes. 153 Football Beyond Borders participants were statistically matched with individuals drawn from the Manchester BeeWell dataset. They found that participants' wellbeing improved by 0.15 points on a 0–10 life-satisfaction scale[33] and they assumed the effects lasted for one year.

During the 2022-23 school year, 2,401 students benefited from the Football Beyond Borders programme, representing a 360 WELLBY benefit. It cost £2.5 million to run the programme. This results in a cost of £6,953 ($8,691)[34] per WELLBY. We consider this a medium-depth analysis.

Happier Lives Institute

The other evaluators we discussed above were not, to our knowledge, explicitly attempting to find the best ways to increase global wellbeing. That, however, has been the goal of the Happier Lives Institute, a research institute we established five years ago to try and answer this question. This led us, based on the reasoning we set out in the introduction, to look for opportunities that are potentially highly cost-effective due to some combination of (1) working on problems that may be very bad for wellbeing, (2) there is an effective intervention, (3) the intervention is very cheaply delivered.

As part of a five-year project, we have estimated the cost-effectiveness, with hugely varying degrees of depth, across 12 charities (10 of which are used in the distribution of results in the core text – the other two were novel analyses). We separate our discussion into evaluations where:

(A) there is a pre-existing published report

(B) we rely on an unpublished report

(C) we undertook novel analyses for the purposes of this WHR chapter.

Those in (A) are more in-depth than those in (B), which are more in-depth than those in (C).

If you are reading this a year or more after the publication of this chapter, it is possible that some of these estimates have changed, or that new evaluations have been produced. For the most up-to-date figures, see: https://www.happierlivesinstitute.org/research/charity-evaluations

Note that the Happier Lives Institute presents results both in 'cost per WELLBY' and 'WELLBYs created per $1,000 donated'. As stated in the main text, these are different ways of presenting the same information. Cost per WELLBY is *cost / effect* whereas WELLBYs created per $1,000 donated is *(effect / cost) *1 000*.

Analyses supported by separate, published reports

We provide more details of our published analyses. These are also presented in order of most to least in-depth (as opposed to say, expected cost-effectiveness)

StrongMinds and Friendship Bench (scaling psychotherapy in low-income countries)

Note this description is identical to the one in the main text.

StrongMinds[35] (in Uganda and Zambia) and Friendship Bench[36] (in Zimbabwe) are two charities scaling access to basic mental healthcare in Sub-Saharan Africa using lay practitioners to deliver psychotherapy to people with depression or anxiety[37].

To estimate the effect of these charities we combined evidence from several sources. For the effect on the direct recipient, we drew on three types of evidence. First, we performed a systematic search and collected 84 randomised control trials (RCTs) of psychotherapy delivered in low- and middle-income countries. Second, we used RCTs associated with the charity's programme (1 for StrongMinds; 4 for Friendship Bench). Third, we used the monitoring and evaluating pre-post data collected by the charities themselves. We then combined the estimates from these different evidence sources based on our judgement of the relative statistical uncertainty, quality, and relevance of each estimate. To estimate the effect on other members of the household, we combined data from six studies[38]. We do not take the effects at face value but adjust them for validity concerns (e.g., adjust for publication bias, see endnote for more detail[39]).

Based on these evidence sources and our methodology, we estimated an overall effect (includes effects over time and impact on the beneficiary's household) of 0.80 WELLBYs for Friendship Bench and 1.80 WELLBYs for StrongMinds per person treated.

The cost of these programmes is very low as they work in low-income countries, and use trained volunteers. The cost to treat an individual for therapy is $17 through Friendship Bench and $45 for StrongMinds. This results in a cost per WELLBY of $21 for Friendship Bench and $25 for StrongMinds. We consider this an in-depth analysis.

GiveDirectly Cash Transfers

GiveDirectly is an organisation that delivers cash transfers to people experiencing poverty. Building on our meta-analysis of 45 RCTs of cash transfers (of all kinds)[40], we meta-analysed the effects of lump-sum cash transfers based on 12 causal studies (n = 24,027, including studies directly about GiveDirectly)[41]. As a result, we estimate that GiveDirectly transferring $1,000 in a lump sum to people in LMICs leads to an increase of 2 WELLBYs for the direct recipient. This is a ~0.4 points effect on a 0–10 scale that decreases over time and the effects end after a decade. Note these effects over time are estimated with data, rather than assumed (as is standard in the non-HLI evaluators).

We also estimate the household effects by synthesising evidence of household effects from 9 causal studies (n = 35,961)[42]. We estimate that the effect on additional household members is about as large as the effect on the direct recipient (86% of the size), and that the average household includes 5 other members than the direct recipient. This increases the total WELLBY effect to 9 WELLBYs.

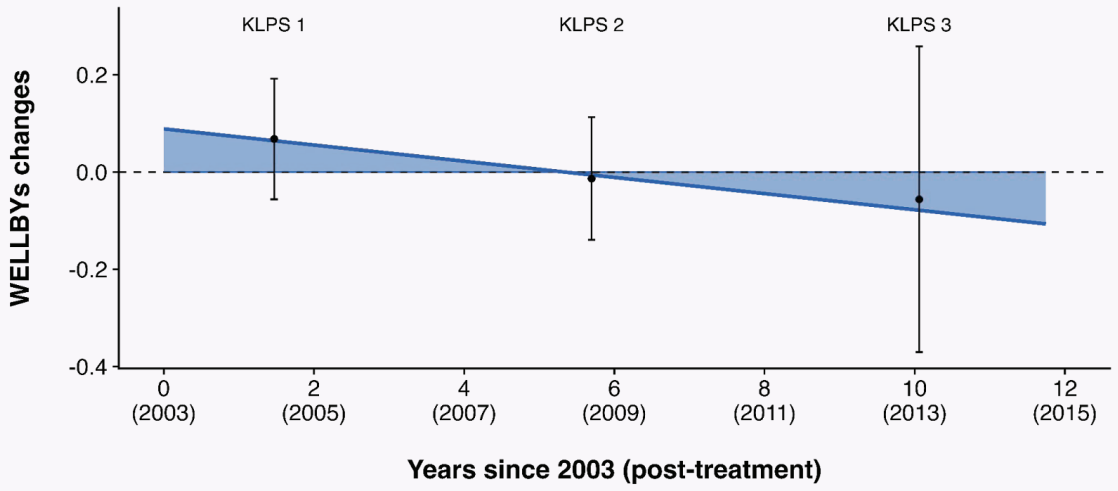
While this is a large effect, given that the cost per cash transfer for GiveDirectly is relatively large (it's ~$1,221 in total to provide a $1,000 transfer, as there are also organisational overheads). This leads to a cost per WELLBY of $132. This is an in-depth analysis. Comparing a $1,000 cash transfer (which costs $1,222 per intervention) vs psychotherapy (which costs $17-$45 per intervention), cash is more effective *per intervention*, but considerably more costly *per intervention* then. Psychotherapy ends up being about 5x more cost-effective.

Note that psychotherapy is being delivered to poor people in low-income countries diagnosed with depression, whereas cash is being delivered to poor people in low-income countries. This is **not** a comparison of taking the average person in absolute poverty, and then providing therapy or cash. We expect that offering therapy to the average person in global poverty (many of whom won't be depressed) would be much less effective than providing it to those diagnosed with depression.

Mass deworming (various charities)

GiveWell, another research-based charity evaluator (which doesn't use WELLBYs) has directed over $163 million to deworming charities since 2010 based on its expected cost-effectiveness in terms of increasing incomes[43]. We undertook what we believe to be the first analysis of deworming's benefits in terms of wellbeing. We analysed happiness data from the long-term follow-up to a deworming RCT in the Kenyan Life Panel Survey (n = 5,094, the KLPS)[44]. We found that deworming has a small, statistically

Figure A1: The effect of mass deworming on wellbeing



non-significant effect on long-term happiness that seems (surprisingly) to become negative over time, see Figure A1.[45]

Given the non-significance, in combination with the trend towards negative effects over time, we do not think it is sensible to estimate the cost-effectiveness of mass deworming from a wellbeing perspective. This is why we do not include it in our graphs in the core text.

However, if we were forced to do so, the (cost) effectiveness would vary radically depending on how we integrate the effects over time. Optimistically, assuming the effects do not become negative (and so overlooking the data) that would lead to a cost per WELLBY of $25. This would be reasonably cost-effective if true because while the effects per person would be tiny – it is incredibly cheap to provide mass deworming. The cost per year of a child dewormed for Deworm the World (a leading charity in the area), is $0.92[46].

However, as one can see in Figure A1, the effects become non-significantly negative over time. We are uncertain what could explain this negative effect, and think it is more likely that we just can't conclude that there are long-term benefits of deworming on wellbeing from this data. Nevertheless, if we took this at face value, and integrated the effects over time over the period shown in this graph, it would result in an overall negative effect on wellbeing. That is for a given investment of dollars it generates negative WELLBYs, or harms, and it would be better not to spend the money on anything. While it may be true that deworming has a positive effect (and is cost-effective), this is not supported by the available data. As such, for donors who want to engage in cost-effective, *evidence-based* giving, it is not a promising option. It could become so, but this would require new data (with a positive result).

We discuss these complex topics of non-significant and uncertain results in much more depth in the report[47]. We consider this an in-depth analysis.

Community Management of Acute Malnutrition by Taimaka

Community Management of Acute Malnutrition (CMAM) is a type of programme which treats cases of acute malnutrition in an outpatient setting with ready-to-use therapeutic foods

(RUTF) to replenish energy and nutrients and by addressing complications (i.e., infections) related to malnutrition. It also provides inpatient care for cases with complications (e.g., severe malaria). Taimaka[48] is an organisation working in Northern Nigeria to deliver CMAM that can treat a child with malnutrition for around $87.

To estimate[49] the wellbeing effect of treating acute malnutrition, we used causal evidence from three related topics to create four estimates, which we then averaged across (see endnote for more detail[50]). We estimate Taimaka's life-improving cost-effectiveness, after discounts, at a cost per WELLBY of $15 (but it varies depending on the evidence source). Taimaka's benefits could be even higher if we considered its life-saving impacts, but we do not present this here. We only briefly mention calculations for life-saving benefits in another analysis, that of lifeboats (see novel analyses at the end of this appendix).

Compared to our most detailed analyses, we rate the depth of work gone into creating this estimate as medium, and the evidence quality as very weak. We consider our work here as highly speculative based on the life-improving effects. Despite the fact we describe these as 'weak' and 'very weak', we should note this level of analysis is substantially more informative about cost-effectiveness than intuitive judgements about how much good a charity does.

The Network for Empowerment & Progressive Initiative (NEPI)

NEPI[51] is attempting to scale the combined use of a cognitive behavioural therapy (CBT) and cash transfer intervention targeting young men of a violent and criminal persuasion to reduce their anti-social activities. This intervention, which NEPI co-developed, was found to have large 10-year effects on criminality (34 fewer thefts a year) and small effects on the mental wellbeing of its recipients (0.19 SDs, equivalent to 0.4 LS points)[52]. NEPI reports that this intervention costs $630 per person treated ($300 for the cash transfer and the rest for therapy and overhead costs).

For our central estimate, we combine the effect from only the direct recipient and their household (3 WELLBYs), and the wellbeing from individuals who are no longer victims of crime (11 WELLBYs), for a total of 14 WELLBYs. This results in a cost per WELLBY of $46. But this improves to $10 if we consider the speculative but plausible alternative analytical choices (e.g., benefits of crime reduction on the community).

We rate the depth of work as medium, but the evidence quality as weak to moderate. We consider our work here as speculative.

Reducing Lead Exposure through advocacy by Pure Earth

Lead is a heavy metal that is toxic to humans. It can cause neurodevelopmental problems in children and impact their future wellbeing. Pure Earth[53] is an organisation whose programmes include advocating for, and lending technical support to, governments in LMICs to create or enforce regulations that preclude the use of lead in common household products like paint, spices, or even cosmetics.

While the cost-effectiveness of advocacy is more speculative to model than direct delivery interventions (e.g., cash or psychotherapy), Pure Earth has a track record of success that make us think that this is potentially a very cost-effective way to avert suffering and increase wellbeing. In our shallow review[54] of lead exposure as a cause area, we reviewed several possible interventions to decrease exposure to lead and concluded that advocacy appeared the most promising. We use longitudinal cohort information about the link between childhood blood lead levels and adult wellbeing. This allows us to model the wellbeing gained per mg/dL of lead in the blood of children prevented. Because this prevents developmental harms that can have long-term effects into adulthood, we are assuming gains over the lifespan. Then, for the advocacy interventions, we predict how successful they will be at reducing blood lead levels and what would have happened without advocacy (i.e., the counterfactual).

In a recent report, we modelled Pure Earth's programme to remove lead from cosmetics in Ghana. For example, Chilo (often known as 'Kohl'), a type of eyeliner applied to both girls and boys, even at a very young age, is very prevalent in northern Ghana[55]. Chilo is often found to have elevated levels of lead which can be ingested through hand-to-mouth contact or absorbed through the skin. This programme has a cost per WELLBY of $9.

Compared to our most detailed analyses, we rate the depth of work that went into creating this estimate as medium and the evidence quality as weak. We consider our work here as speculative. We note there are other organisations that work on lead exposure, such as the Lead Exposure Elimination Project (LEEP)[56]. We have not (yet) undertaken side-by-side analyses here.

Early childhood psychosocial stimulation

The International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b), traditionally a research institute, is trying to scale the Reach Up parenting programme in Bangladesh[57]. The Reach Up programme teaches and encourages parents to play with their children in a more stimulating and compassionate way.

There is some evidence that this programme has sizable effects on mental wellbeing (0.4 SDs) nearly thirty years later[58]. Using this evidence along with other evidence of psychosocial stimulation's long-term effects (a total of 2 RCTs in LMICs, n = 426; 4 causal studies in HICs, n = 2,502), we estimate that early psychosocial stimulation has a total effect of 5 WELLBYs per person (effects are integrated over 32 years)[59]. We also estimate that it costs the icddr,b $98 per child treated, leading to a cost per WELLBY of $20.

Compared to our most detailed analyses, we rate the depth of work that went into creating this estimate as medium and the evidence quality as weak. We consider our work here as speculative.

Fortifying Wheat with Iron by Fortify Health

Fortify Health[60] is a charity working in India that persuades mills to add iron to their flour (fortification) and provides technical support to do so. It costs them $0.03 to supply a year's worth of wheat flour. We estimate that this has a small effect on mental wellbeing (equivalent to 0.02 LS points per year of provision) through two steps[61]. First, iron fortification of wheat decreases anaemia by 23% (5 RCTs, n = 2,315)[62]. Anaemia (of which around half the population of India suffers) is associated with an increased likelihood of depression (e.g., in a meta-analysis of 14 studies, anaemia is associated with a 40% increase in the likelihood of depression in the general population[63]) and depression is deeply related to subjective wellbeing.

Our analysis suggests that the cost per WELLBY through iron fortification of wheat by Fortify Health is $46.

Compared to our most detailed analyses, we rate the depth of work that went into creating this estimate as medium and the evidence quality as weak. We consider our work here as speculative.

Unpublished analyses reported in this analysis

There are two unpublished analyses which we include in our review of WELLBY cost-effectiveness analyses of charities. We explain their calculations here.

Improved flooring by Earthenable

EarthEnable[64] is a charity-for-profit hybrid where the charity oversees the operation of several for-profits in East Africa that build earthen floors that they claim[65] are comparable in effectiveness but much cheaper to install and repair than concrete floors. The most recent data we have suggests a cost of $300 per household treated[66] for a household provided with an earthen floor.

There is no estimate of the effect of earthen floors on subjective wellbeing. However, Cattaneo et al. (2009; n = 2,742) using a natural experimental design evaluated the effect of a government programme in Mexico to build cement floors, 2-4 years after it occurred. They found an effect equivalent to 0.34 life satisfaction points (0 to 10) for the adult respondents around three years later. We assume the effects will persist as long as cash transfers do (8.6 years) and have similar effects on the household as cash transfers[67]. This is because many cash transfer recipients spend it

on housing improvements like metal roofs[68]. The resulting total effect is around 8.8 WELLBYs, which leads to a cost per WELLBY of $34 to improve flooring with earthen floors through Earthenable.

Compared to our most detailed analyses, we rate the depth of work that went into creating this estimate as very low and the evidence uality as moderate. We consider our work here as very speculative.

Increasing emergency housing for individuals living in poverty with TECHO

TECHO[69] is an organisation known for building transitional homes called mediaguas for people living in slums in Latin America. In 2021, they built 1,820 emergency homes and around 300 more permanent dwellings. They now do many things including water access and education but to build an emergency TECHO emergency house costs around $1000, and their overhead seems to be 29%[70], implying a total cost per house of around $1,290.

Galiani et al. (2018) and Mitchell et al. (2019) studied the effects of a lottery-designed trial (basically an RCT) to provide TECHO emergency housing (TECHO) to people living in inadequate housing and slums in El Salvador, Uruguay, Mexico and Argentina (n = 3,018), at an average of 15 months after the house was delivered.

Galiani et al. (2018) estimated the duration of the benefits. They found that the effects decay considerably after a further 8 months and predicted that the wellbeing effects decay to nothing by 28 months. Assuming the trend Galiani estimates is correct, this would imply a total benefit to the recipient of 4 WELLBYs spread across 2.3 years, which if shared across the average Latin American household (4.6; Pew, 2020), would represent a large effect of 4 * 4.6 = 18.4 WELLBYs. Altogether this would lead to a cost per WELLBY of $70.

Compared to our most detailed analyses, we rate the depth of work that went into creating this estimate as very low and the evidence quality as moderate. We consider our work here as very speculative.

Novel analyses reported in this analysis

In the chapter, we performed several novel back-of-the-envelope calculations to estimate the cost-effectiveness of more well-known charitable funding opportunities.

Royal National Lifeboat Institution

The Royal National Lifeboat Institution (RNLI)[71] is a long-running and beloved organisation primarily dedicated to the hard and often dangerous work of trying to rescue people in danger at sea off the coast of the UK. In 2023, they report saving a total of 269 people[72] and a total expenditure of £242.6m[73] (around $303.3m)[74]. These figures imply a cost of $902k per life saved.

RNLI have other benefits than saving lives, but we will assume the benefit is primarily the lives they saved. We comment on this briefly below.

Many researchers and decision-makers view the value of a life saved as the happy years it gains a person. To illustrate the value of saving a life we adapt the depiction given in Layard and Oparina (2021):

$$Future\ wellbeing = \sum_t [Wellbeing_t - Neutral\ point] \cdot (1 - \delta)^t$$

Where:

- Wellbeing represents the reported well-being of the individual at year t.
- The neutral point represents where wellbeing shifts from negative to positive on the wellbeing scale (e.g., 5 on a 0 to 10 scale).
- δ is the discount rate, meant to capture fundamental uncertainty about the future.
- The summation is done over the expected lifespan the person has left, t.

This isn't the only way to conceptualise the value of saving a life, so we don't think it's appropriate to naively compare life-saving to life-improving charities. But this formulation suffices to allow us to compare the wellbeing effect between life-saving charities.

See Plant et al. (2022) for a discussion of the difficulties and hard choices faced by decision-makers when deciding how to assign

a wellbeing value to averting a death. Note that there are philosophically justified alternatives that would mean a lower cost-effectiveness.

In this case, we assume the typical age of a person saved is the average age in the UK (40.7)[75] and the life expectancy in the UK is (78.6)[76] according to the Office of National Statistics (ONS). This means that there are on average 38 years of life saved per person saved. The average wellbeing in the UK in 2023 was 6.66 the 0–10 Cantril Ladder[77]. We use a neutral point of 2, but this is based on limited data from Samuelsson et al. (2023) and two unpublished studies by Tessa Peasgood and Julian Jamison. Note that this is the same value used in HM Treasury's Green Book and is based on the unpublished study by Tessa Peasgood. We are loathed to put so much weight on so little data of questionable relevance, but a neutral point of 2 is the closest there is to a 'standard' figure. The result is that each year of life saved will produce 4.66 points of wellbeing. We do not apply a discount rate.

Altogether this implies the average value of a life saved is 177 WELLBYs, and 47,489 WELLBYs for all the lives saved. This results in a cost per WELLBY of $6,386.

Arguably, this undercounts the RNLI's value, as it provides a psychological benefit to those who don't directly use its service, but could do – the feeling that someone would save you if you need it. However, any attempt to estimate the size of this benefit would be very speculative. Further, many charities also have a "it's good to know it's there if you need it" factor, so it's not obvious how accounting for this factor would change the numbers across the board. Hence, we don't try to account for it here. We thank the editors for bringing this issue to our attention.

This analysis is very shallow. We can confidently assess the cost-per-output (the cost of rescuing someone) but not the value-per-output (the value of rescuing someone).

Homeless cash transfers

In the chapter, we discussed cost-effectiveness analyses for homelessness interventions. This includes our shallow estimation of the cost-effectiveness of cash transfers for the homeless in Canada, repeated below.

We considered one study looking at cash transfers to homeless people living in Vancouver Canada[78]. This RCT looked at unconditional 7,500 CAD (6,637 USD) lump-sum cash transfers to homeless people in Canada, so a large sum. Based on the results of the trial, we model that it has an impact lasting for two years, leading to an overall effect of 0.33 WELLBYs, at a cost of $6,667 per person[79]. This leads to a cost per WELLBY of $19,994. This is more promising than 'housing first', but still much less impactful per dollar than the best-performing alternatives.

Guide Dogs

The Guide Dogs for the Blind Association spent around £74m on guide dogs in 2023[80] and reported that they formed 469 new guide dog partnerships. We estimate, based on three limited studies, that having a guide dog leads to a 0.7 life satisfaction points increase (on a 0 to 10)[81] – which is a very large effect per year. Guide dogs work on average 7 years[82]. Therefore, we estimate that one extra guide dog leads to 4.83 WELLBYs. Note, we are just counting the impact on humans. It's impact for 2023 is 2,266 WELLBYs, which means a cost per WELLBY of $40,767.

Endnotes

1 See our methodology for more detail: https://www.happierlivesinstitute.org/research/cost-effectiveness-analysis-methodology/.

2 In Dupret et al. (2024) we tested how similar results from our different evaluations are on typical subjective wellbeing scales (like life satisfaction) and on affective mental health scales (like depression). We found little difference in results between these broad types of scales. If anything, results on subjective wellbeing tended to be slightly (but not significantly) higher than results on affective mental health. Therefore, adding affective mental health scales is unlikely to overestimate results, so we do not apply an adjustment. We add these measures because there still is a dearth of results on typical subjective wellbeing scales, especially for interventions in low-income countries. This is a first step in exploring this topic, and more research on conversion between scales is welcome.

3 https://actionforhappiness.org/.

4 In Frijters and Krekel (2021) but based on an RCT by Krekel et al. (2021).

5 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

6 https://www.royalvoluntaryservice.org.uk/.

7 Krekel et al. (2024).

8 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

9 https://humanhenge.org/about/.

10 In Frijters and Krekel (2021) but based on a pre-post study described in Heaslip and Darvill (2017) and Drysdale (2018).

11 Converted from results on the WEMWBS scale (Frijters & Krekel, 2021).

12 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

13 This was based on Stergiopoulos et al. (2015), a well-powered (n = 2,148) trial in Canada that compared the effect of housing and social assistance to the typical assistance provided. They estimated the costs based on studies of similar interventions in the UK. See Frijters and Krekel (2021, p. 210) for their discussion of it.

14 Frijters and Krekel (2021).

15 https://www.stateoflife.org/.

16 State of Life has also performed at least two other CEAs of government supported programmes: The National Citizen Service 2015 summer programme (to encourage youth to volunteer), which found a cost per WELLBY of $3,157 (State of Life, 2017), and the Tideway Legacy Programme (an infrastructure project to expand London's sewer system), which found a cost per WELLBY of $11,655.42 (State of Life, 2023b).

17 By relevant we mean that the surveys involve the direct participants in the charity's interventions. This of course, a comparative term and its use is in comparing how generalizable different sources of evidence will be. Naturally, you can't do better in terms of relevance (i.e., generalizability or external validity), than evidence about the intervention as it's practised by the charity at scale.

18 They describe the project as "Put simply, a CCT process is a Bible-based guided training approach that a local church can use with its community to identify the resources they have – working together to create long-lasting, whole-life change. As local communities work through the transformational change to communities and process, they decide what outcomes and activities work best in their situation. For example, this might be running training on vocational skills, establishing community projects, or setting up support and savings groups." (State of Life, 2023c, p. 1).

19 State of Life (2023c).

20 "Therefore, in order to avoid overstating the benefits of CCT and presenting implausibly high benefit-cost ratios, we report our main Net Social Benefit (NSB) and Social BCR using direct benefits only on one hand and total costs (both direct and indirect) on the other." (State of Life, 2023c). This would lead to a cost-effectiveness of $2 per WELLBY.

21 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

22 https://www.londonyouthrowing.com/.

23 https://heyzine.com/flip-book/907baf2538.html.

24 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

25 https://www.probonoeconomics.com/Listing/Category/about-us.

26 https://walkingwiththewounded.org.uk/.

27 Pro Bono Economics (2022).

28 This was employment, mental health difficulties, age, marital status, ethnicity, sex, health conditions, and job status (Pro Bono Economics, 2022, p. 28).

29 They note missing data for those who didn't complete the programmes, low response rates (45% and 32%) for the life-satisfaction question, small sample sizes (around ~100), and they also mention that there may be bias due to the parallel trends assumption being violated (e.g., there is regression to the mean in the Walking with the Wounded sample that is not present in the matched data).

30 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

31 Pro Bono Economics (Franklin, 2024) and Cheng and Humphrey (n.d).

32 https://www.footballbeyondborders.org/.

33 Pro Bono Economics (Franklin, 2024; see also Cheng & Humphrey (n.d) used the effect on students considered to be 'at risk'. They found an improvement of 2.4 points on the SWEMWBS measure of mental wellbeing, which they convert at a 0.24 rate to life satisfaction on a 0–10 scale; namely, an effect of 0.576. They assume benefits only occur for the 'at risk' children. They adjust the impact by the proportion of 'at risk' student in the total sample of Football Beyond Borders in the 2022-2023 school year (26%), resulting in an average effect per student of 0.26*0.576 = 0.15 points.

34 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

35 https://strongminds.org/.

36 https://www.friendshipbenchzimbabwe.org/.

37 McGuire et al. (2024b).

38 Note that there is no spillover data directly related to the charities programme. We use wider sources of data to estimate a spillover ratio (5 RCTs and 1 controlled trial). Namely, we estimate that non-recipient members of the household experience 16% of the wellbeing benefit that the psychotherapy recipient experiences. We apply this to the average household in the countries where these charities operate, which is between 3 and 4 other household members.

39 We discounted this estimate for a range of internal validity (i.e., is the data accurate) and external validity (i.e., does the data we use generalise to the case we are estimating) concerns. For example, for internal validity, we multiply the effect by 0.69 (a 31% discount) for publication bias. For example, for external validity, we multiply the effect by 0.90 (a 10% discount) for the use of lay therapists. The charities rely on lay deliverers of manualised psychotherapy instead of experts because there are too few experts in low income countries. This is often called 'task-shifting'. While our modelling suggests it reduces the effect a little bit, it also allows for a lot more people to be treated and at lower costs; hence, it improves the cost-effectiveness of the charities.

40 McGuire (2020; 2022a).

41 McGuire and Plant (2021a, 2021d).

42 McGuire et al. (2022b).

43 https://docs.google.com/spreadsheets/d/1wCI-auzETS2q0m_I-9YJboPMMuoNy1bMO3yWIE_mWsU/.

44 Hamory et al. (2021).

45 Dupret et al. (2022).

46 See Appendix A.1.2 of Dupret et al. (2022).

47 Dupret et al. (2022).

48 https://taimaka.org/.

49 McGuire et al. (2024c).

50 Specifically, we extrapolated our estimates from the long-term effects of: famine on wellbeing, a protein intervention in Guatemala on wellbeing and small-quantity lipid-based nutrient supplements (SQ-LNS) on (a) cognition and (b) socio-emotional skills. We then roughly convert these outcomes to wellbeing, and use the pathway with the larger effect as our estimate. The estimated total impact of CMAM in WELLBYs from the three sources of evidence is broadly similar, and also include large adjustments (aka discounts): Famine: 6.94 WELLBYs (after a 0.56 adjustment); Protein: 6.45 WELLBYs (after a 0.10 adjustment); SQ-LNS (IQ): 7.17 WELLBYS (after a 0.41 adjustment); SQ-LNS (social-emotional skills): 2.39 WELLBYs (after a 0.41 adjustment).

51 https://www.nepiliberia.org/.

52 Blattman et al. (2023).

53 https://www.pureearth.org/.

54 McGuire et al. (2023b).

55 In a survey, Pure Earth (2024; n = 3,227) found that 93% of children from the Northern part of Ghana used Chilo.

56 https://leadelimination.org/.

57 https://reachupandlearn.com/reach-up-in-bangladesh/.

58 Walker et al. (2022).

59 McGuire et al. (2024e).

60 https://www.fortifyhealth.global/.

61 McGuire et al. (2024d).

62 Field et al. (2021).

63 Lee and Kim (2020).

64 https://www.earthenable.org.

65 https://earthenable.org/hm/about-us-2/#:~:text=What%20Is%20The-,Evidence,-That%20Earthen%20Floors.

66 https://www.earthenable.org/wp-content/uploads/2022/06/EarthEnable-Consolidated-Financials-for-year-2021-Unaudited.pdf.

67 McGuire et al. (2022b).

68 See Tables B.1 and B.2 of Haushofer and Shapiro (2016).

69 https://techo.org/rendicion-de-cuentas/.

70 See Slide 24 of https://issuu.com/techo_org/docs/memoria_techo_int._2021_1_.

71 https://rnli.org/.

72 See the RNLI's (2023) annual report.

73 See the 2023 expense information from the Charity Commission for England and Wales.

74 Charities evaluated in the UK generally have results reported in pound sterling. For consistency, we convert these results to USD, using the average conversion rate reported by the World Bank (2023): $1 = £0.8, so we convert results with £X * 1/0.8.

75 https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/bulletins/annualmidyearpopulationestimates/mid2022#:~:text=The%20median%20age%20of%20the%20population%20in%20the%20UK%20was,median%20age%20at%2043%20years.

76 https://www.ons.gov.uk/peoplepopulationandcommunity/birthsdeathsandmarriages/lifeexpectancies/bulletins/nationallifetablesunitedkingdom/2020to2022.

77 See the data from World Happiness Report 2024.

78 Dwyer et al. (2023).

79 This cost consists of the $5,555 USD cash transfer and assuming it costs 20% of the value of the transfer to deliver it. We assume this figure because it's a similar value to what we estimated as the overhead share of the cost of a cash transfer delivered by GiveDirectly, a well run organization (McGuire & Plant, 2021a).

80 See Guide Dogs' (2023) annual report.

81 We take a sample-weighted average of Refson et al. (1999, Scotland, effect is 0.80 on a 0–10 scale converted linearly from the SWLS; n = 117), Yarmolkevich (2017, USA, effect is 1.08 on a 0–10 scale converted linearly from the SWLS; n = 58), and Glenk et al. (2019, Austria, effect is -0.30 on a 0–10 scale converted linearly from the WHOQOL-BREF psychological subscale, n = 36). These studies are comparing visually impaired individuals with and without guide dogs, they are associative and not causal. We do not apply an adjustment for the lack of causality. This is a very shallow analysis.

82 See Guide Dogs' FAQ.